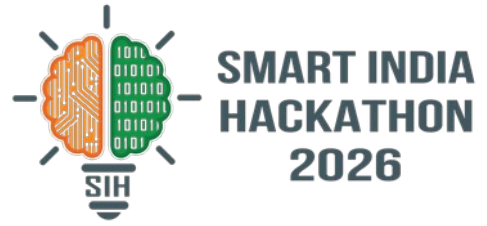


SMART INDIA HACKATHON 2026



KrishiSetu

Problem Statement ID– **SIH26033**

Problem Statement Title –**Farmer-first agricultural trading platform with pooled delivery**

Theme – **Agriculture, FoodTech & Rural Development**

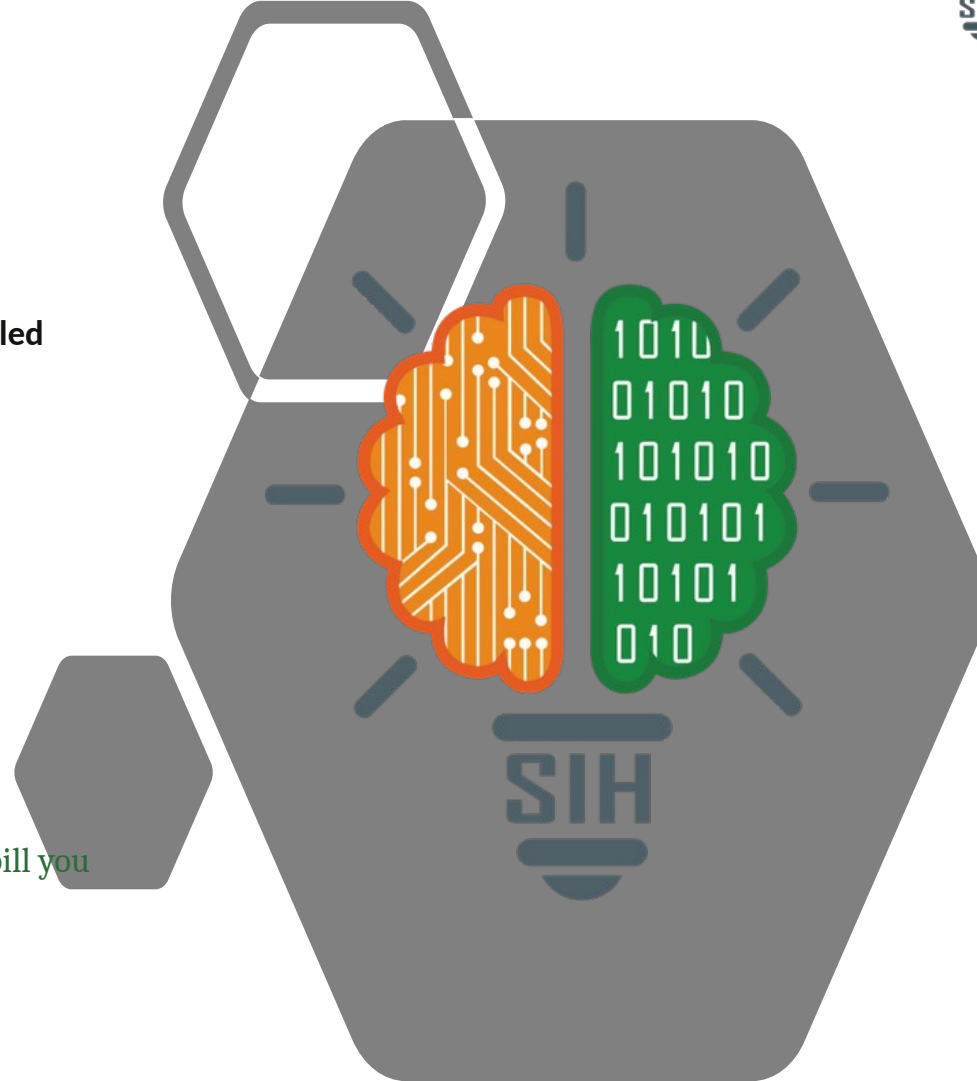
PS Category – **Software**

Team ID – **SIH26-SW059**

Team Name – **Logic_Lords — Ramrao Adik Institute of Technology**

The farmer names their price. Everything after that is shared transport and a bill you can read.

Working prototype live at krishisetu-weld.vercel.app



KRISHISETU —

POOLED FARMER

1 KG TOMATO • BUILT UPWARD



₹45.66
buyer pays

vs



₹55.00
quick-commerce reference

BUYER ORDER DETAILS

ONE 5 KG BUYER ORDER, FILLED FROM TWO NAMED FARMS



Sanjay Patil

Khalapur, Raigad



3.0 kg

₹102.00



Meena Bhoir

Panvel Rural, Raigad



2.0 kg

₹68.00



Worked with the same pricing function the live site uses. ₹55/kg is an assumed quick-commerce reference, not a live scrape.



What is Actually Different?

1

Farmer gets the net rate



100% Farmer Rate

No commission, listing charge or logistics cost is deducted from anywhere.

2

Orders are pooled by destination



Only lots going to the same neighbourhood share a vehicle.

3

Nothing moves unsold



The tempo is loaded against confirmed orders, so there is no stock to write off.

4

Better price, healthier food



Shorter chain, fresher produce, more value for everyone.



How the Buyer Receives It?

A



Cluster pickup point

Kirana or society office holds the batch, 6–9 pm window.



B



Last 300 metres

Hand cart inside the same society, free above a ₹500 basket.



C



Express, on demand

Not offered — it cannot pay for itself at a 6 kg basket.



Same quality



Less middlemen



Lower cost



Fresher food



TECHNICAL APPROACH

Our working name for the decision layer. It is a product name for our own engine, not a claim of a new published algorithm.



8 KEY CONSIDERATIONS



BUILD & RUN



RULES THE CODE ENFORCES



Routing returns a feasible nearest-neighbour plan compared against a fixed-order baseline over the same stops. We do not claim a global optimum.

FEASIBILITY AND VIABILITY

RETURN LOADS – WHY THE EMPTY HALF OF A TRIP MATTERS



A transporter who brings vegetables into the city drives home with an empty vehicle, and that wasted half is already priced into what he charges us. A return load is any cargo he can carry on the way back — seed, fertiliser, packaging, etc. FPO's supplies — matched to the corridor and time window he has already committed to. He accepts on deciding it himself, and the quoted payment must still cover the real detour: empty space does not automatically mean a cheaper trip.



123
orders pooled



747.7 kg
on a 750 kg tempo



5
stops: 4 farms, 1 drop



₹5.37
transport per kg

RISK, AND WHAT IS BUILT AGAINST IT



Too few orders in one locality

The run is held below 70% fill and rolls to the next window. A public invite page shows neighbours the exact shortfall.



Produce spoils in transit

Freshness risk and counter-shipment commodity; an over-age lot is never loaded on an unrefrigerated run.



Quality disputes

Declared grade, pre-stop handover records, and a resolution that can only deduct from a farmer with recorded agreement.



Farmer cannot use an app

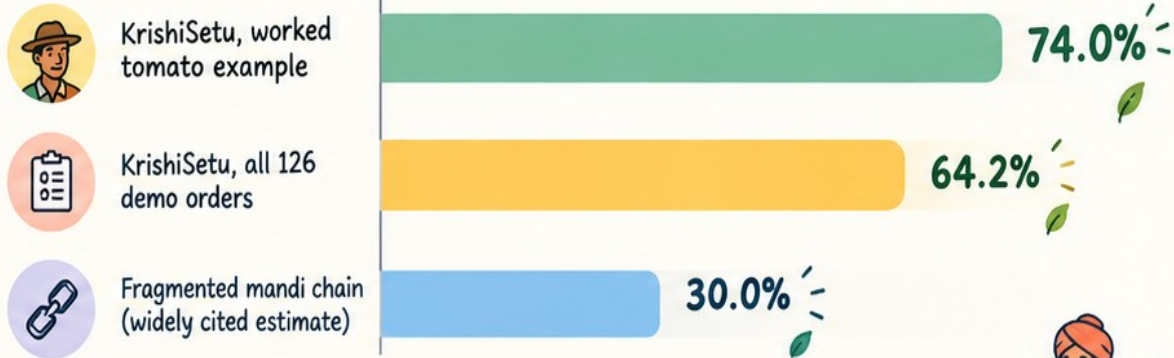
Marathi, Hindi and English on the farmer screens; one decision card answers what to send and what it pays.



Return-load matching is designed and specified; it is deliberately after the core proof, and is not claimed as shipped.

IMPACT AND BENEFITS

Farmer's share of what the buyer pays (%)



The fragmented-chain figure is a widely reported range for Indian vegetables, shown as an estimate. The two KrishiSetu bars are computed from orders in the running system.



Where the platform's own money comes from

	Disclosed buyer-side fee per order	₹19.89
	Payment gateway, messaging, support	– ₹5.60
	Contribution per completed order	₹14.29
	Assumed monthly fixed cost	₹60,000
	Orders per month to break even	4,197

No farmer listing fee. No commission on settlement. When a price must be held under the quick-commerce reference, the site fee is cut to zero first – the farmer's accepted amount is never touched.

Farmer

₹34.00/kg accepted and paid in full, besides a deleted mandi comparison

Buyer

₹45.66/kg against a ₹55 reference, every bill line itemised

Transporter

₹4,016 for one full run, agreed before acceptance

Platform

₹14.29 per order, entirely from a disclosed buyer-side fee

Transparent • Fair • Sustainable

Figures are computed over synthetic demonstration orders, so they size the model rather than measure a business. No production cost is recorded, so no column here is farming profit.

RESEARCH AND REFERENCES

Sources, and what they do and don't prove

- 01**  **Government of India Open Data Portal**

Live prices of essential commodities and market trends.

data.gov.in/catalog/current-daily-price-various-commodities-various-market-mandi


- 02**  **eNAM – trading and FPO pages**

Details on electronic trading and Farmer Producer Organisations (FPO).

enam.gov.in/web/trading-details


- 03**  **Ninjacart and Ninja Mandi**

B2B vegetable supply chain and mandi operations.

ninjacart.com/ninja-mandi


- 04**  **agribazaar**

Marketplace, logistics and warehousing claims.

agribazaar.com/quick-links/steps


- 05**  **Samunnati**

FPO financing and market linkage.

samunnati.com/innovations


- 06**  **Google OR Tools, YRP with fine windows**

Reference for simulation for capacitated routing with windows.

developers.google.com/optimization/routing/vrptw


- 07**  **PID release on agricultural marketing**

Government position on market reform.

pib.gov.in → PIB/222802


- 08**  **SIH 2026 idea submission format**

Template this deck follows.

sih.gov.in/letters/2026



What this does not prove: demand history and the fitted price series in the prototype are synthetic and labelled so on screen. An unmentioned capability on a competitor's public page is not evidence they lack it, and we claim no measured superiority without a comparable pilot.